

→ ML Data Pre-processing

- Handling missing data
- Handling outliers
- Feature scaling
- Encoding variable

Missing values occur when data is not recorded for a variable.

NaN
None
Empty
Special Symbol

$$IQR = Q3 - Q1$$

1-hot ~~encoding~~ encoding
1-hot decoding

Student Academic Dataset

Student ID	Study Hr/Day	Program	CGPA	Scholarship
S1	4	CSE	3.5	Yes (1)
S2	NaN	BBA	3.8	Yes (1)
S3	5	NaN	3.2	No (0)
S4	6	CSE	NaN	Yes (1)
S5	3	EEE	3.0	No (0)

→ Replace with mean =

$$\frac{4+5+6+3}{4} = 4.5$$

 → Replace with med =
 3, 4, 5 6

$$\left(\frac{4+5}{2}\right) = 4.5$$

Replacing using Mode (प्रमुख)
 ↓
 "CSE"

→ Replace with mean =

$$\frac{3.5+3.8+3.2+3.0}{4} = 3.375$$

 → Replace with median =
 3.0 3.2, 3.5 3.8

$$\frac{3.2+3.5}{2} = 3.35$$

Target encoding with program

$$CSE = \frac{(Yes + No + Yes)}{3} = \frac{(1+0+1)}{3} = 0.67$$

Table: Website Visitor Data

Day	D1	2	3	4	5	6	7	8	9	10
Visitor	120	135	128	140	145	150	155	160	170	500

Alays sorted data শাক্ষণ না: sort করতে হবে।

Day = D1, D3, D2, D4, D5, D6, D7, D8, D9, D10

Visitor = 120, 128, 135, 140, 145, 150, 155, 160, 170, 500

Number of observations, $n=10$

Splitting in lower and upper bound

120 128 135 140 145 || 150, 155, 160, 170, 500

Find Q_1 and Q_3

lower half = 120 128 135 140 145

Q_1 = Median of lower half = 135

upper half = 150, 155, 160, 170, 500

Q_3 = Median of upper half = 160

$IQR = Q_3 - Q_1 = 160 - 135$

Outlier boundaries

$$LB = Q_1 - 1.5 \times IQR$$

$$= 135 - (1.5 \times 25)$$

$$= 97.5$$

$$HB = Q_3 + 1.5 \times IQR$$

$$= 197.5$$

Any value less than 97.5 or greater than 197.5 is an outlier
 $\therefore 500 > 197.5 \therefore$ it is an outlier.

hasan | ③

→ Why handling missing values is important

- Incorrect model training
- Biased ...

→ Common reasons for missing data.

→ Types of missing values

① MCAR (Missing completely at Random)

② MAR (Missing at Random)

③ MNAR (Missing NOT at Random)

→ Handling missing data

→ Feature Scaling

① Normalization

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

② Standardization (Score Scaling)

$$X' = \frac{X - \mu}{\sigma}$$

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Encoding Categorical Variable

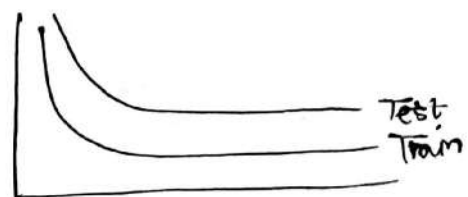
1. वास्तविक संख्या

PDF → ML Data Preprocessing

→ Weighted Target Encoding

Train	Test	Validation
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4. Data Leakage



Shayan / ④

[5.] K-Fold Target Encoding